Semester One Examination, 2025

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

SOLUTIONS

Section Two: Calculator-assumed

WA student number: In figures

In words

Your name

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Number of additional answer booklets used (if applicable):

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.

3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.

4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

5. It is recommended that you do not use pencil, except in diagrams.

6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed **65% (98 Marks)**

This section has **twelve** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 8 **(7 marks)**

The universal set U contains the 26 uppercase letters of the alphabet and sets X, Y and Z are

$$X = \{P, R, O, F, U, S, E\}, \qquad Y = \{C, H, A, L, K, Y\}, \qquad Z = \{P, N, E, U, M, O, G, A, S, T, R, I, C\}$$

(a) Determine the set $X \cap Z$. (1 mark)

Solution
$X \cap Z = \{P, R, O, U, S, E\}$
Specific behaviours
✓ correct set, including curly braces

(b) Use set notation to describe the set of letters that are in set Y or not in set Z. (1 mark)

Solution
$Y \cup \bar{Z}$ or $Y \cup Z'$
Specific behaviours
✓ correct set notation

(c) State, with justification, which two sets are mutually exclusive. (2 marks)

Solution
X and Y are mutually exclusive since $X \cap Y$ is the empty set.
Specific behaviours
✓ correct pair of sets ✓ any correct explanation/justification

(d) Determine

(i) $n(\overline{X \cap Z})$. (1 mark)

(ii) $P(X \cup Y \cup Z)$. (2 mark)

Solution
$n(\overline{X \cap Z}) = 26 - 6 = 20$ $P(X \cup Y \cup Z) = \dfrac{13 + 4 + 1}{26} = \dfrac{18}{26} = \dfrac{9}{13}$
Specific behaviours
✓ 20 ✓ 18 ✓ simplified

AO ok

End of questions

Question 9 **(7 marks)**

(a) State the equation of the line of symmetry of the graph of $y = 4x^2 - 32x + 51$. (1 mark)

Solution
$x = -\dfrac{-32}{2(4)} \to x = 4$
Specific behaviours
✓ correct equation

(b) State the roots of the graph of $y = -2(x+1)(x-3)(x-8)$. (1 mark)

Solution
$x = -1, \quad x = 3, \quad x = 8$
Specific behaviours
✓ correct roots

(c) Let $f(x) = 2\sqrt{x+4}$ for $0 \le x < 32$. Determine the range of $f(x)$. (2 marks)

Solution
$f(0) = 4, f(32) = 12$ Range is $4 \le y < 12$ or $y \in [4,12)$.
Specific behaviours
✓ correct lower bound and inequality ✓ correct upper bound and inequality

(d) The function $g(x) = ax^2 + bx + c$ has a turning point at $(5, -8)$ and passes through the point $(2, 10)$. Determine the value of each of the constants a, b and c. (3 marks)

Solution
Turning point form: $y = a(x-5)^2 - 8$ Using $(2, 10)$: $10 = a(2-5)^2 - 8 \Rightarrow 9a = 18 \Rightarrow a = 2$ Expanding: $y = 2(x-5)^2 - 8 = 2x^2 - 20x + 42$ $a = 2, \quad b = -20, \quad c = 42$
Specific behaviours
✓ correct turning point form of equation, with constant ✓ correct value of a ✓ correct values for b and c

Question 10 **(9 marks)**

Isaiah creates a biased die with uneven odds to use to win at games. The frequency of each integer generated during a testing session was recorded in the table below and then used to calculate the relative frequency.

Integer	1	2	3	4	5	6
Frequency	30	28	15	42	**44**	41
Relative frequency	0.15	0.14	0.075	0.21	**0.22**	0.205

(a) Show that 200 rolls were conducted during the testing session. (1 mark)

Solution
Let n be the number generated and then $30 \div n = 0.15 \Rightarrow n = 30 \div 0.15 = 200$
Specific behaviours
✓ divides any frequency by its relative frequency

(b) Complete the two missing entries in the table above. (2 marks)

Solution
Missing frequency: $200 - (30 + 28 + 15 + 42 + 41) = 44$ Missing relative frequency: $44 \div 200 = 0.22$
Specific behaviours
✓ missing relative frequency ✓ missing frequency

(c) Use the testing session data to determine the probability that

(i) the next integer generated will be greater than 2. (2 marks)

Solution
$p = 1 - (0.15 + 0.14) = 0.71$
Specific behaviours
✓ writes correct expression for probability ✓ evaluates expression

AO ok

(ii) the next integer generated will be 3, given that it is less than 4. (2 marks)

Solution
$p = \dfrac{P(3 \cap < 4)}{P(< 4)} = \dfrac{0.075}{0.15 + 0.14 + 0.075} = \dfrac{0.075}{0.365} = \dfrac{15}{73} \approx 0.2055$
Specific behaviours
✓ writes correct expression for probability ✓ correct probability

(iii) the next two integers generated will not both be 2. (2 marks)

Solution
$\mathrm{P(Both\ are\ 2)} = 0.14 \times 0.14 = 0.0196$ $p = 1 - 0.0196 = 0.9804 = \dfrac{2451}{2500}$
Specific behaviours
✓ correct probability for both 2 ✓ correct probability

AO ok

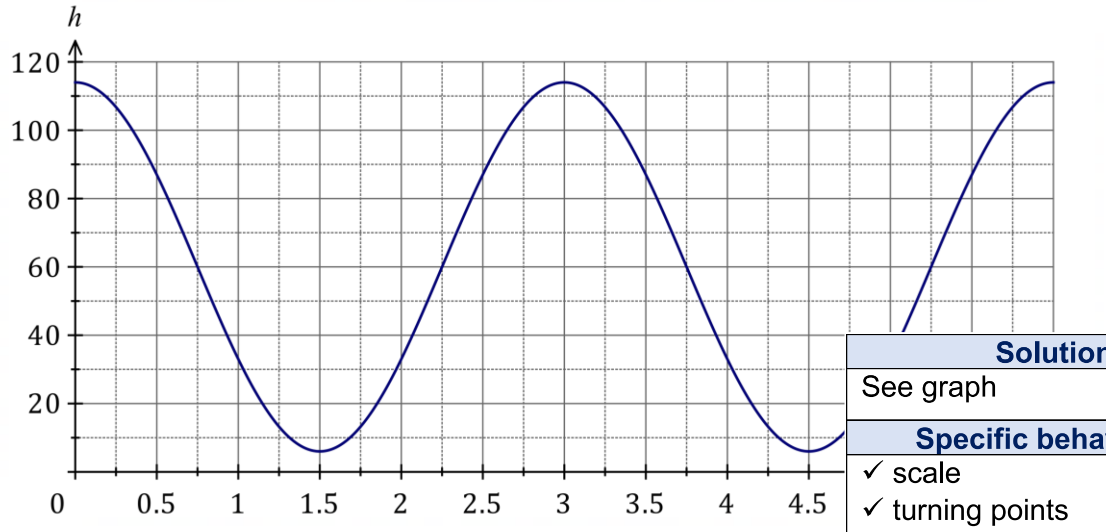
End of questions

Question 11 **(9 marks)**

Jiyeong goes bungee jumping and moves in a vertical line so that at time t seconds, her height h metres above the ground is given by

$$h = 60 + 54\cos\left(\frac{2\pi t}{3}\right)$$

(a) By first adding a suitable scale to the vertical axis, sketch the graph of h against t for $0 \le t \le 6$. (4 marks)



Solution
See graph
Specific behaviours
✓ scale
✓ turning points
✓ points where $h = 60$
✓ reasonable curve

The closest that Jiyeong ever comes to the ground is d metres.

(b) State the value of d. (1 mark)

Solution
$d = 60 - 54 = 6$
Specific behaviours
✓ correct value

(c) Determine the number of times that Jiyeong is d metres above the ground during the first minute. (2 marks)

Solution
The period is 3 seconds and so during the first minute there will be $60 \div 3 = 20$ complete cycles - the number of times will be 20.
Specific behaviours
✓ indicates period of motion
✓ correct number of times

(d) Determine the total time for which Jiyeong is at least 87 metres above the ground between $t = 0$ and $t = 4$. (2 marks)

Solution
$h = 87 \Rightarrow t = 0.5, 2.5, 3.5$ At least 87 m above ground when $0 \le t \le 0.5$ and when $2.5 \le t \le 3.5$, a total time of $0.5 + 1 = 1.5$ seconds.
Specific behaviours
✓ indicates times when $h = 87$
✓ correct length of time with correct units **(UNITS QUESTION)**

Question 12 **(9 marks)**

(a) Evaluate the number of combinations given by $\binom{33}{3}$. (1 mark)

Solution
$\binom{33}{3} = 5456$
Specific behaviours
✓ correct value

(b) State the value(s) of n and the value(s) of r when $\binom{n}{r} = \dfrac{n!}{8!\,5!}$. (2 marks)

Solution
$n = 8 + 5 = 13$ $r = 5$ or $r = 8$
Specific behaviours
✓ value of n ✓ values of r

Sophia can only fit 3 desserts on a single plate at a buffet. So, Sophia selects 3 desserts from 13 desserts offered at the buffet, 7 of which are hot and the remainder cold.

(c) How many different selections are possible? (2 marks)

Solution
$\binom{13}{3} = 286$
Specific behaviours
✓ indicates use of combination formula   ✓ correct number

AO ok

(d) How many different selections are possible if they must all be cold? (2 marks)

Solution
$\binom{13-7}{3} = \binom{6}{3} = 20$
Specific behaviours
✓ use of combination formula with restriction   ✓ correct number

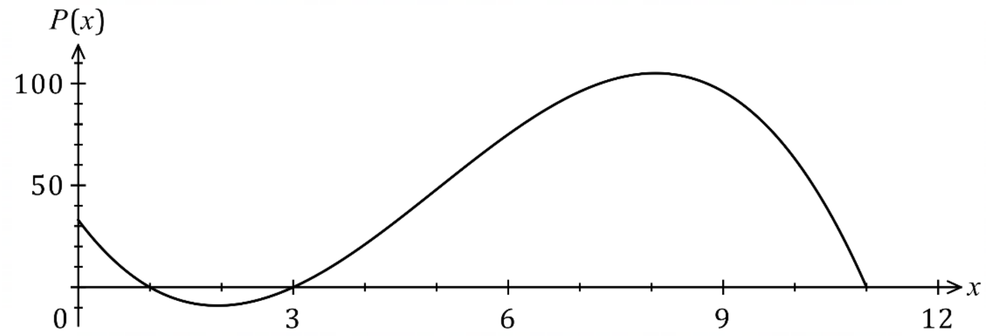
(e) If the selection is made at random, determine the probability that all the desserts chosen are hot. (2 marks)

Solution
$\binom{7}{3} = 35$ $\mathrm{P(All\ hot)} = \dfrac{35}{286} \approx 0.1224$
Specific behaviours
✓ correct number   ✓ correct probability

End of questions

Question 13 **(7 marks)**

The function defined by $P(x) = 33 - 47x + 15x^2 - x^3$ for $0 \le x \le 11$ was used to model the monthly profit P, in thousands of dollars, when a factory set the production rate of a chemical to x thousand litres per day. The graph of $y = P(x)$ is shown.



(a) Determine $P(2.7)$ and interpret your answer in the context of the question. (2 marks)

Solution
$P(2.7) = -4.233$ When the rate of production is 2700 litres per day, the factory will make a loss of $\$4233$ per month.
Specific behaviours
✓ correct value of function ✓ correct interpretation

(b) Express $P(x)$ in factored form. (1 mark)

Solution
$P(x) = -(x-1)(x-3)(x-11)$
Specific behaviours
✓ correct factored form

(c) State the range of the function for the given domain of $0 \le x \le 11$. (1 mark)

Solution
$-9.0276 \le y \le 105.0276$ or $x \in [-9.0276, 105.0276]$
Specific behaviours
✓ correct range

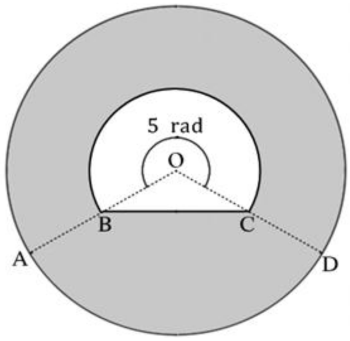
(d) By only considering production rates in the interval $0 \le x \le 8$, determine the change in the production rate required to increase the monthly profit by $\$30\,000$ from its value when the factory is producing 4400 litres per day. (3 marks)

Solution
$P(4.4) = 31.416$ $P(x) = 31.416 + 30 = 61.416 \Rightarrow x = 5.483$ The rate must increase by $5483 - 4400 = 1083$ litres per day.
Specific behaviours
✓ evaluates current monthly profit ✓ forms equation for required profit and solves for x ✓ correct increase in rate

Question 14 **(8 marks)**

The diagram shows a circle with a hole through the middle.

$\angle AOD = 5$ radians and
$AB = BO = OC = CD = 3.3$ cm



(a) Determine the perimeter of the shaded region to 2d.p. (4 marks)

Solution
Circumference of bigger circle $2\pi(6.6) = 41.47$ Major arc length of smaller circle $5 \times 3.3 = 16.5$ $BC = \sqrt{3.3^2 + 3.3^2 - 2 \times 3.3 \times 3.3 \times \cos(2\pi - 5)} = 3.95$ Perimeter is $41.47 + 16.5 + 3.95 \approx 61.92$ cm
Specific behaviours
✓ correct expression for outer circle ✓ correct expression for arc length ✓ correct expression for BC ✓ correct perimeter

(b) Determine the area of the shaded region to 2d.p. (4 marks)

Solution	
Big circle area $\pi \times 6.6^2 = 136.848$ Small circle area $\pi \times 3.3^2 = 34.212$ Area of small circle minor segment $\frac{1}{2}(3.3)^2\left((2\pi - 5) - \sin(2\pi - 5)\right) = 1.7656$ Area is $136.848 - 34.212 + 1.7656 \approx$ $104.40cm^2$	Big circle area $\pi \times 6.6^2 = 136.848$ Area of small circle major sector $\frac{1}{2}(3.3)^2 \times 5 = 27.225$ Area of ΔOBC $\frac{1}{2}(3.3)^2 \times \sin(2\pi - 5) = 5.221$ Area is $136.848 - 27.225 - 5.221 \approx$ $104.40cm^2$
Specific behaviours	
✓ correct expression for big circle area ✓ correct expression for small circle area ✓ correct expression for small circle minor segment ✓ correct shaded area	✓ correct expression for big circle area ✓ correct expression for small circle major sector ✓ correct expression for ΔOBC ✓ correct shaded area

-1 for rounding error on page
ROUNDING QUESTION

End of questions

Question 15 **(9 marks)**

A group of Rossmoyne students are asked whether they study Economics or History. Analysis of the 399 responses showed that 57 study both, 190 study neither and 171 study Economics.

(a) Determine how many of these students study History but not Economics. (2 marks)

Solution
$n(H \cap \bar{E}) = 399 - 171 - 190 = 38$
Specific behaviours
✓ indicates any correct method ✓ correct number

AO ok

(b) One of the students is chosen at random. Determine the probability that they

(i) study Economics or History. (2 marks)

Solution
$n(E \cup H) = 399 - 190 = 209$ $\mathrm{P}(E \cup H) = \dfrac{209}{399} = \dfrac{11}{21} \approx 0.5238$
Specific behaviours
✓ correct $n(E \cup H)$ ✓ correct probability

(ii) study History given that they study Economics. (2 marks)

Solution
$\mathrm{P}(H\|E) = \dfrac{57}{171} = \dfrac{1}{3} = 0.\bar{3}$
Specific behaviours
✓ correct numerator ✓ correct probability

(c) For these students, are the events studying Economics and studying History independent? Justify your answer. (3 marks)

Solution		
$n(H) = 38 + 57 = 95$ $\mathrm{P}(H) = \dfrac{95}{399} = \dfrac{5}{21} \approx 0.2381$ $P(H) \not\approx P(H\|E)$ $0.2381 \not\approx 0.\bar{3}$	$n(E) = 171$ $P(E) = \dfrac{171}{399} = \dfrac{3}{7} \approx 0.4286$ $P(E) \not\approx P(E\|H)$ $0.4286 \not\approx 0.6$	$P(E \cap H) = \dfrac{57}{399} = \dfrac{1}{7}$ $\dfrac{171}{399} \times \dfrac{95}{399} = \dfrac{5}{49} \approx 0.1020$ $P(E) \times P(H) \not\approx \dfrac{1}{7}$ $0.1020 \not\approx 0.1429$
As the calculated probability and actual probability are not similar enough based on the independence test we completed, we conclude that the events are not independent.		
Specific behaviours		
✓ indicates correct test for independence ✓ compares LHS and RHS ✓ statement to conclude that events are not independent		

Question 16 **(8 marks)**

A small tin and a large tin each contain a mixture of green and red plastic counters. There are 7 green and 4 red counters in the small tin, and 6 green and 8 red counters in the large tin.

(a) When two counters are removed at random and without replacement from the large tin, determine the probability that they are both green. (2 marks)

Solution
$\mathrm{P}(G \cap G) = \frac{6}{14} \times \frac{5}{13} = \frac{15}{91} \approx 0.1648$
Specific behaviours
✓ correct probability for first or second pick seen ✓ correct probability

(b) When one counter is removed at random from each tin, determine the probability that the counters are the same colour. (3 marks)

Solution
$\mathrm{P}(G \cap G) = \frac{7}{11} \times \frac{6}{14} = \frac{3}{11}$ $\mathrm{P}(R \cap R) = \frac{4}{11} \times \frac{8}{14} = \frac{16}{77}$ $\mathrm{P}\big((G \cap G) \cup (R \cap R)\big) = \frac{3}{11} + \frac{16}{77} = \frac{37}{77} \approx 0.4805$
Specific behaviours
✓ correct probability for both green ✓ correct probability for both red ✓ correct probability

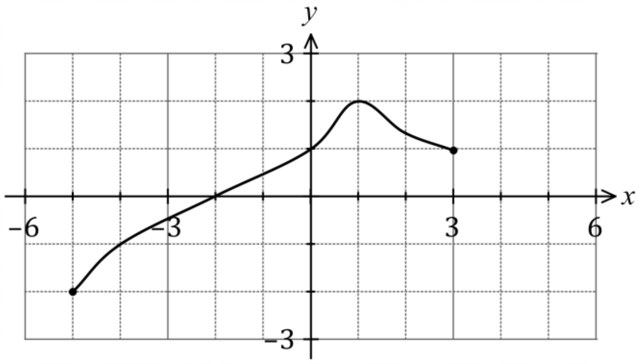
(c) When a counter is removed at random from the large tin and placed in the small tin, determine the probability that the next counter drawn at random from the small tin is red. (3 marks)

Solution
$P(R_S \vert R_L) = \frac{8}{14} \times \frac{5}{12} = \frac{5}{21}$ $P(R_S \vert G_L) = \frac{6}{14} \times \frac{4}{12} = \frac{1}{7}$ $P(R_S \vert R_L \cup R_S \vert G_L) = \frac{5}{21} + \frac{1}{7} = \frac{8}{21} \approx 0.3810$
Specific behaviours
✓ correct probability for red if red taken from large tin ✓ correct probability for red if green taken from large tin ✓ correct probability

End of questions

Question 17 **(8 marks)**

The graph of $y = f(x)$ is shown.



(a) Determine $f(-4)$. (1 mark)

Solution
$f(-4) = -1$
Specific behaviours
✓ correct value

(b) State the domain of the function f. (1 mark)

Solution
$-5 \le x \le 3$ or $x \in [-5, 3]$
Specific behaviours
✓ correct domain

(c) State the range of the function f. (1 mark)

Solution
$-2 \le y \le 2$ or $x \in [-2, 2]$
Specific behaviours
✓ correct range

(d) Solve $f(x) = 1$. (1 mark)

Solution
$x = 0, x = 3$
Specific behaviours
✓ correct solutions

(e) Determine the **coordinates** of the following points:

(i) the x-axis intercept of the graph of $y = f(x - 3)$. (1 mark)

Solution
$(-2 + 3, 0) \rightarrow (1, 0)$
Specific behaviours
✓ correct coordinates

(ii) the y-axis intercept of the graph of $y = -8f(x)$. (1 mark)

Solution
$(0, 1 \times -8) \rightarrow (0, -8)$
Specific behaviours
✓ correct coordinates

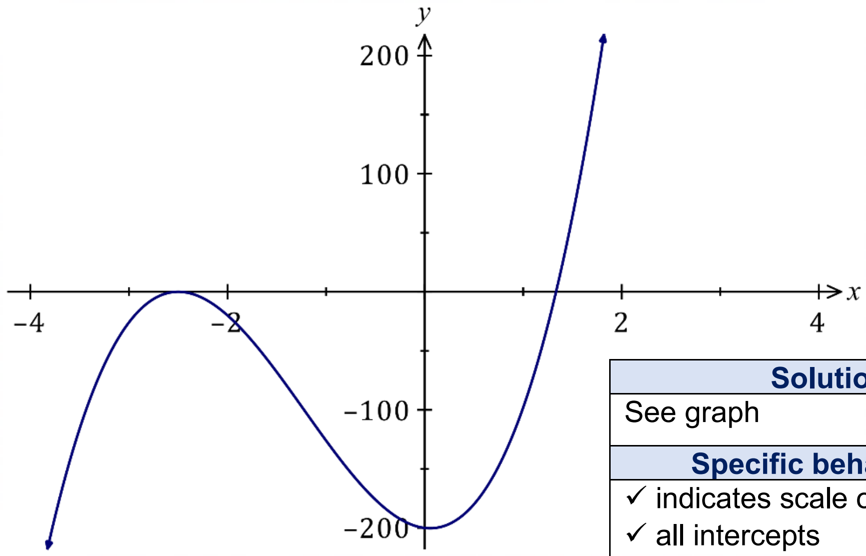
(iii) the turning point of the graph of $y = f(0.5x) - 3$. (2 marks)

Solution
$(1 \div 0.5, 2 - 3) \rightarrow (2, -1)$
Specific behaviours
✓ correct x-coordinate
✓ correct y-coordinate

Question 18 **(7 marks)**

The graph of $y = f(x)$, where f is a cubic polynomial, has a turning point at $\left(-\frac{5}{2}, 0\right)$ and passes through the points with coordinates $(0, -200)$ and $\left(\frac{4}{3}, 0\right)$.

(a) Sketch the graph of $y = f(x)$ on the axes below. (3 marks)



Solution
See graph
Specific behaviours
✓ indicates scale on each axis ✓ all intercepts ✓ cubic curve through points

(b) Determine $f(x)$ in expanded form $ax^3 + bx^2 + cx + d$ where a, b, c and d are constants. (4 marks)

Solution

Using roots:

$$f(x) = a\left(x + \frac{5}{2}\right)^2\left(x - \frac{4}{3}\right)$$

$$f(0) = a\left(0 + \frac{5}{2}\right)^2\left(0 - \frac{4}{3}\right) = -200 \Rightarrow a = 24$$

$$f(x) = 24\left(x + \frac{5}{2}\right)^2\left(x - \frac{4}{3}\right)$$
$$= 24x^3 + 88x^2 - 10x - 200$$

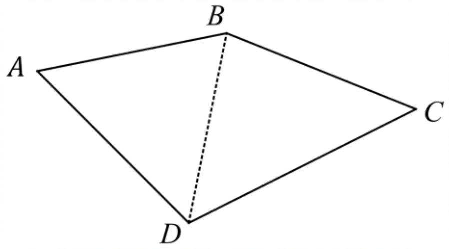
Specific behaviours

✓ attempts to use roots and constant to form f
✓ uses repeated root as quadratic factor
✓ uses y-intercept to evaluate a
✓ correct $f(x)$ in expanded form

End of questions

Question 19 **(10 marks)**

Ann buys a plot of land to start farming. The diagram shows a sketch of the plot of land $ABCD$ with diagonal BD.



Known lengths (in metres) are $CD = 62.5$, $BC = 46.5$ and $BD = 48.3$, and known angles are $\angle ADB = 54°$ and $\angle ABD = 56.5°$.

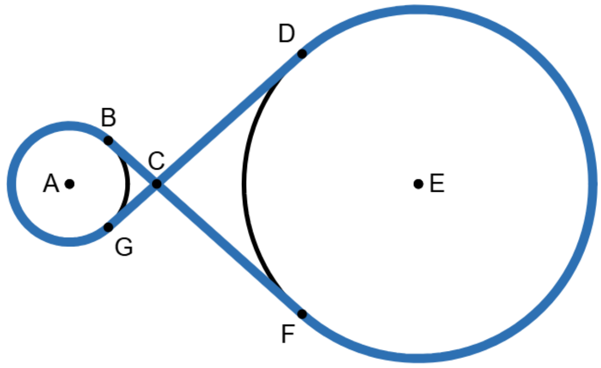
(a) Show use of trigonometry to determine the size of $\angle BCD$. (2 marks)

Solution
$\angle BCD = \cos^{-1}\left(\dfrac{46.5^2 + 62.5^2 - 48.3^2}{2(46.5)(62.5)}\right)$ $= 50.0°$
Specific behaviours
✓ reasonable attempt to use cosine rule   ✓ correct angle

(b) Show use of trigonometry to determine the length of the side AD. (2 marks)

Solution
$\angle A = 180° - 56.5° - 54° = 69.5°$ $\dfrac{AD}{\sin(56.5°)} = \dfrac{48.3}{\sin(69.5°)} \Rightarrow AD = 43.0\text{ m}$
Specific behaviours
✓ reasonable attempt to use sine rule   ✓ correct length

(c) On Ann's farm, there are two water towers that are 1m and 3m in radius, and 2m apart. Ann takes a can of spray paint and draws on the ground, around the two water tanks and connects the paint together as shown in the diagram. The intersection of the paint trail at C is 1.5m away from the centre of the smaller water tower.
What is the length of this paint trail? (6 marks)
Note: Use radian measure



Solution
$AC = 1.5m$ and $EC = 4.5m$ $\angle BAC = \angle GAC = \cos^{-1}\left(\frac{1}{1.5}\right) = 0.841$ $\angle DEC = \angle FEC = \cos^{-1}\left(\frac{3}{4.5}\right) = 0.841$ $2\pi - 2 \times 0.841 = 4.601$ $1 \times 4.6 = 4.601m$ and $3 \times 4.6 = 13.803m$ $CB = CG = \sqrt{1.5^2 - 1^2} = 1.118m$ $CD = CF = \sqrt{4.5^2 - 3^2} = 3.354m$ Total = $4.601 + 13.803 + 2 \times (1.118 + 3.354) = 27.348m$
Specific behaviours
✓ uses 1.5m and 4.5m ✓ uses trigonometric ratios to find angle 0.841 ✓ calculates major arc angle of two circles ✓ calculates major arc lengths of two circles ✓ calculates length of straight lines paths ✓ correct total

End of questions

Supplementary page

Question number: _________

Supplementary page

Question number: _________

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